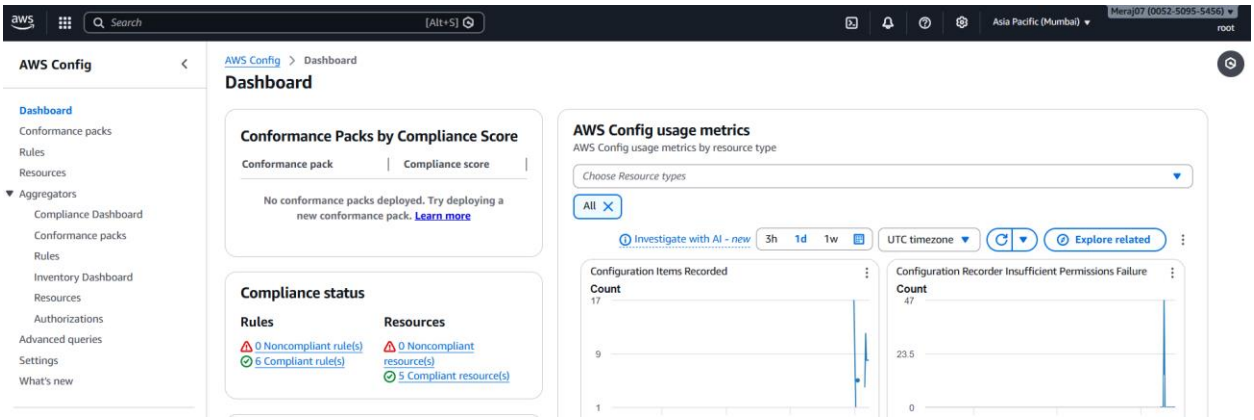
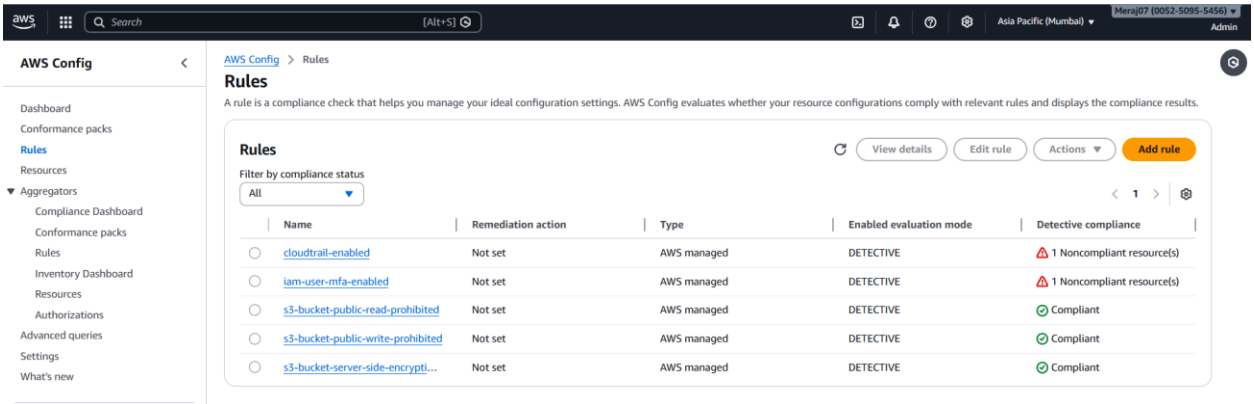
Alerting and remediation flow: CloudTrail → CloudWatch / EventBridge → Lambda → SNS

1. AWS config Dashboard - AWS Config dashboard showing overall compliance status of monitored resources.

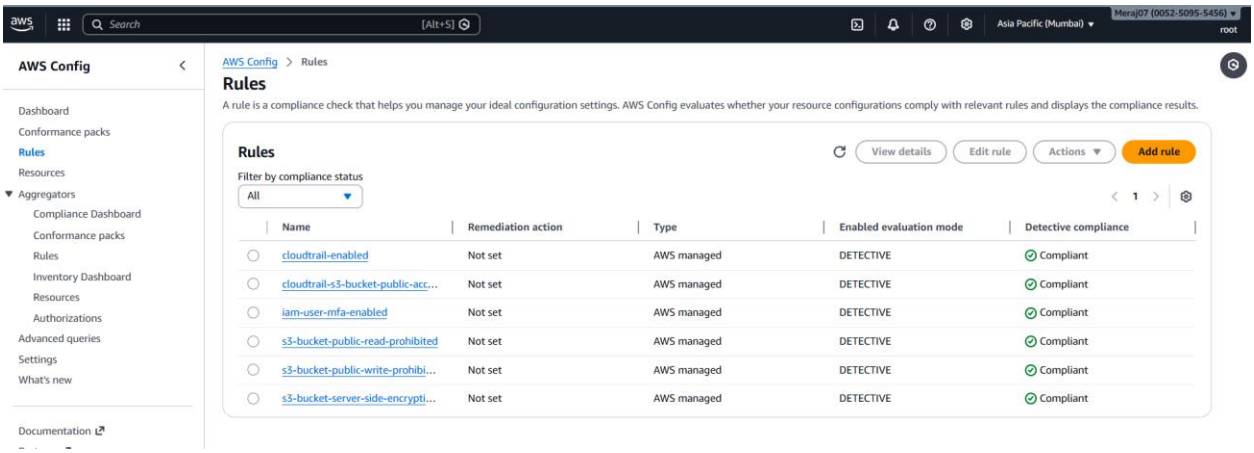


1.1 AWS Config Rules

Before Compliance – Non-compliant resources detected.

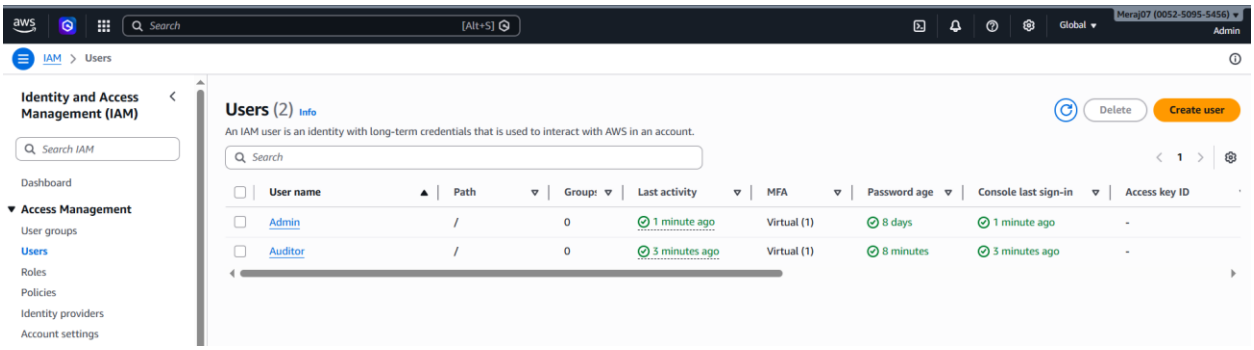


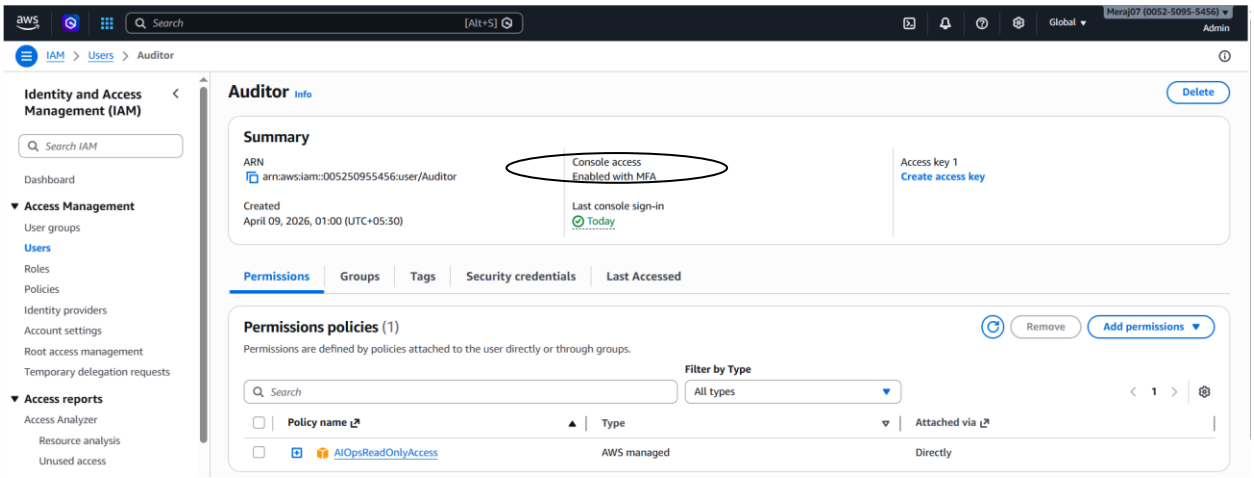
After Compliance – After Compliance – All resources compliant after manual remediation actions.



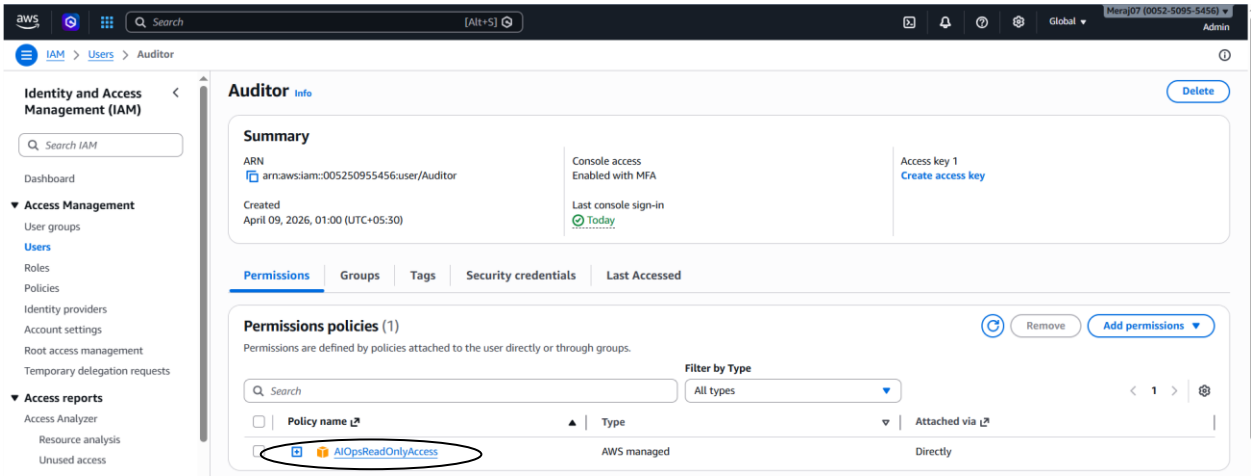
2. IAM & Access Control

IAM users configured with MFA with RBAC

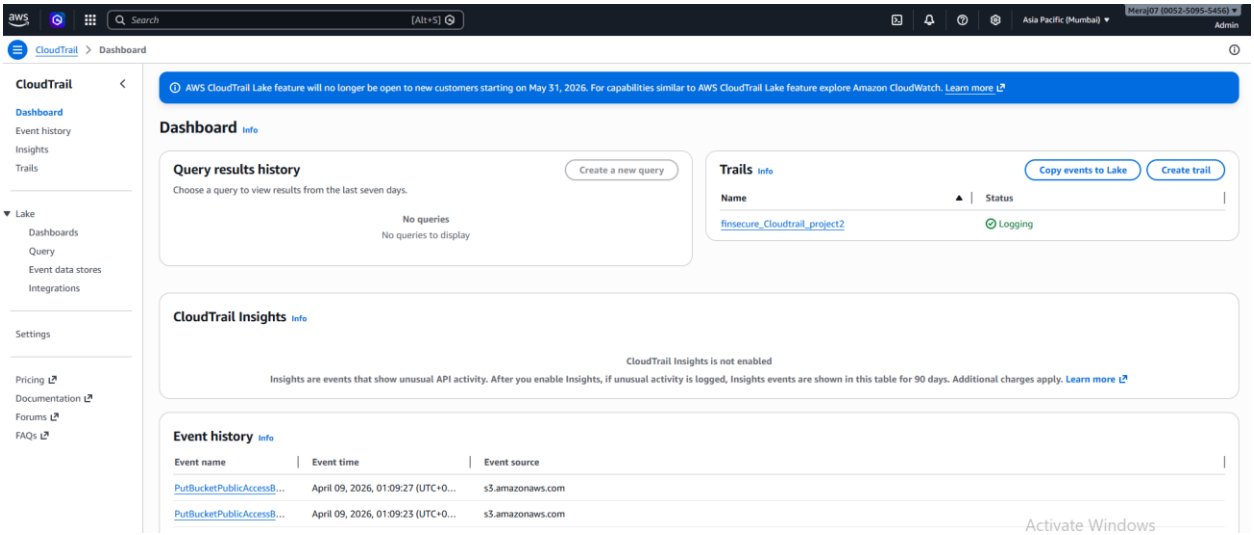




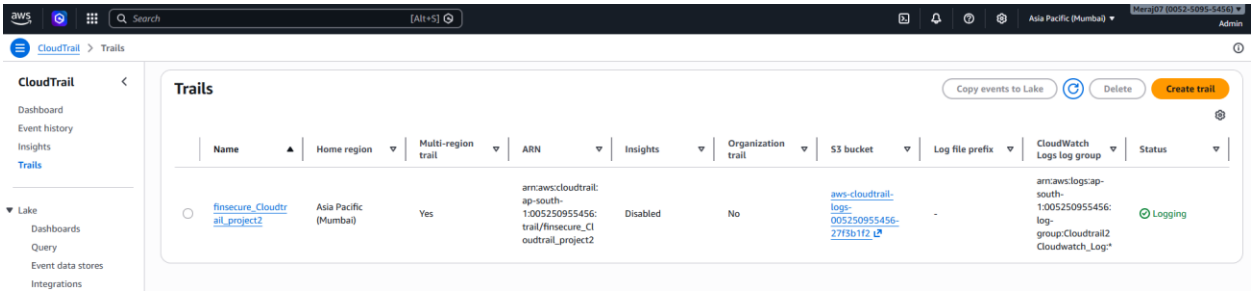
2.1 IAM policies configured with least privilege access for each role.



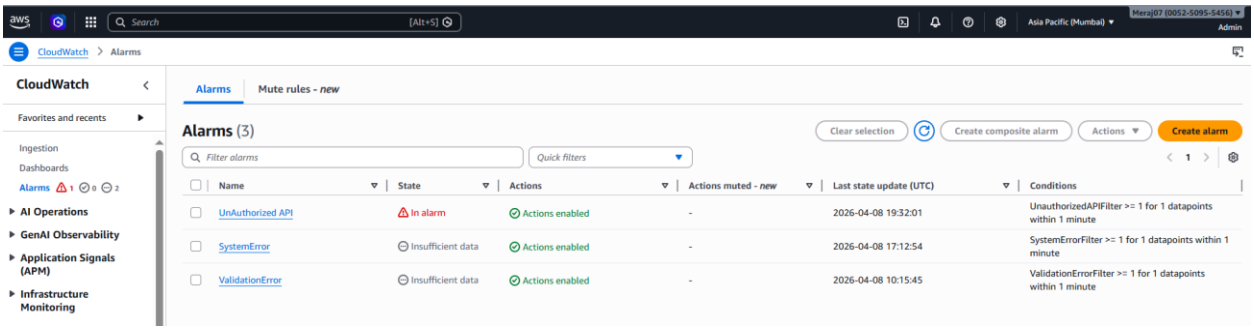
3. CloudTrail Logging - CloudTrail enabled to capture all AWS API activity across services.



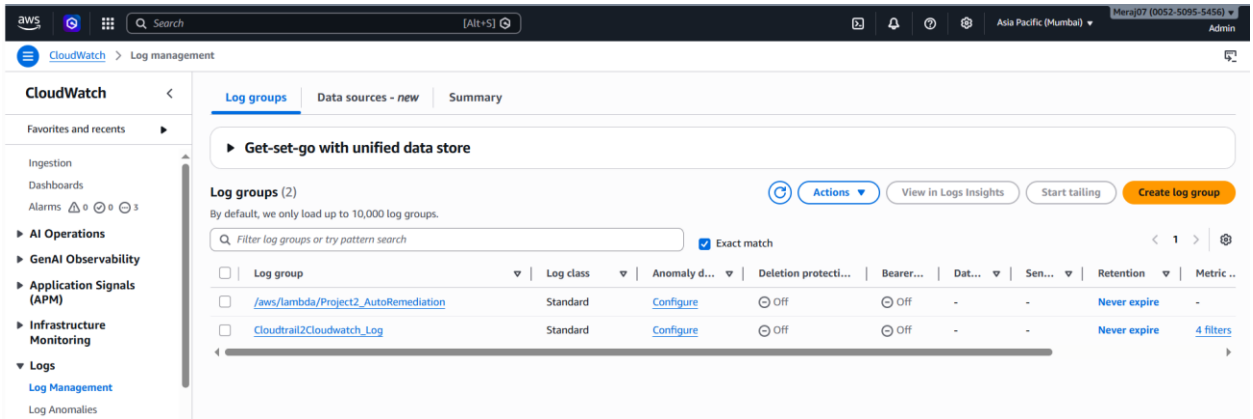
3.1 CloudTrail Multi-region - CloudTrail configured to log events across all AWS regions.



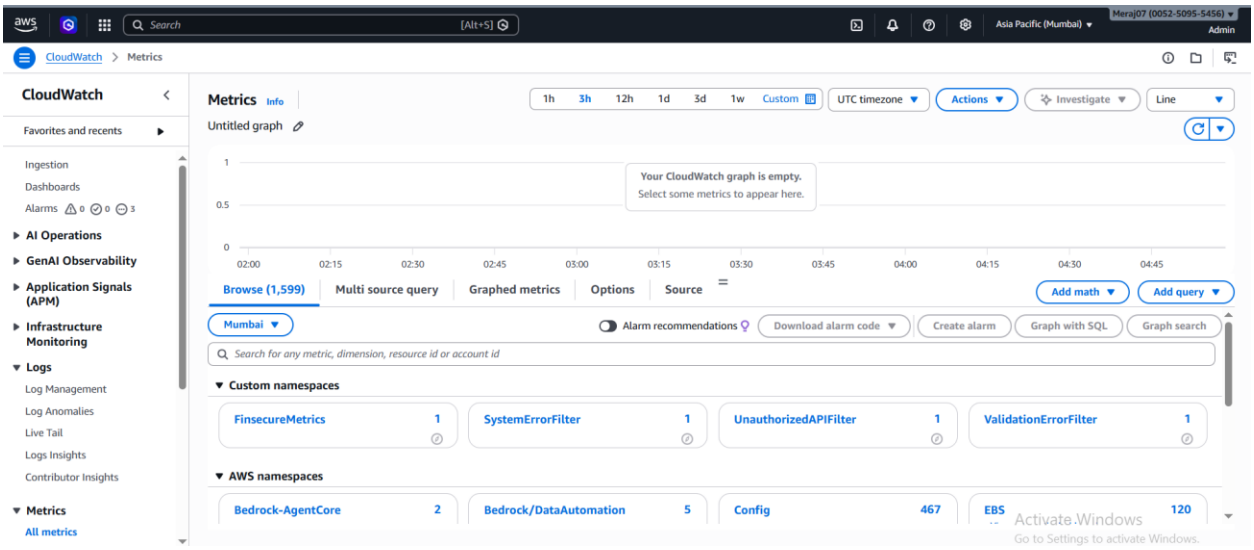
4. CloudWatch Monitoring - CloudWatch used for monitoring logs and triggering alarms on security events.



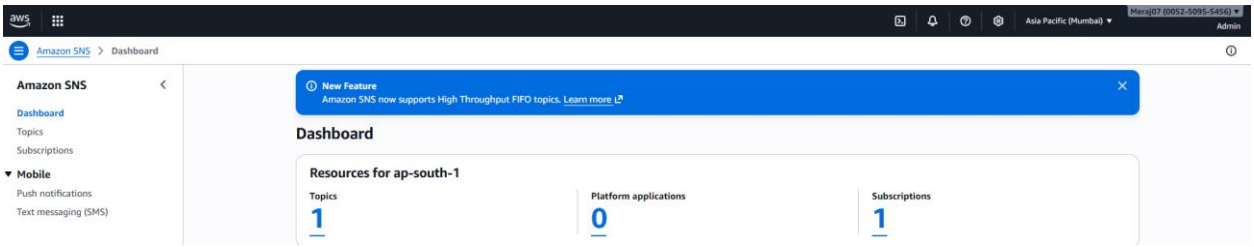
4.1 CloudWatch Logs - CloudTrail logs ingested into CloudWatch Logs for analysis.

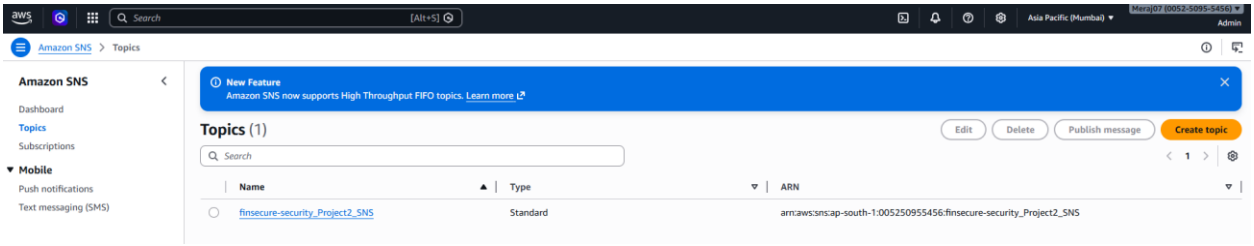


4.2 CloudWatch Metric Filters - Metric filters configured to detect security events like root login and unauthorized API calls.

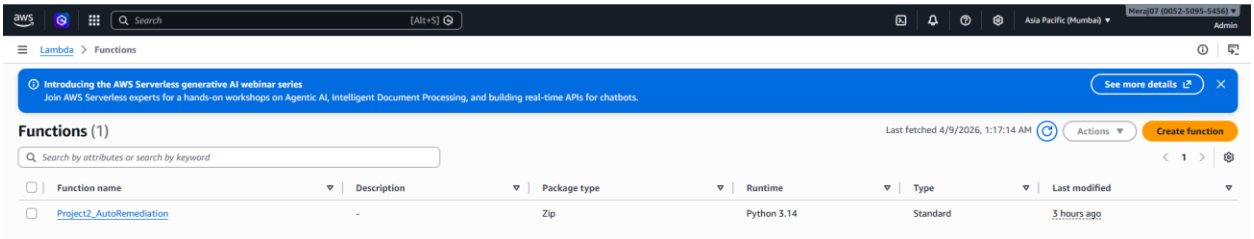


5. SNS Notifications - SNS topic configured to send real-time email alerts for security events.

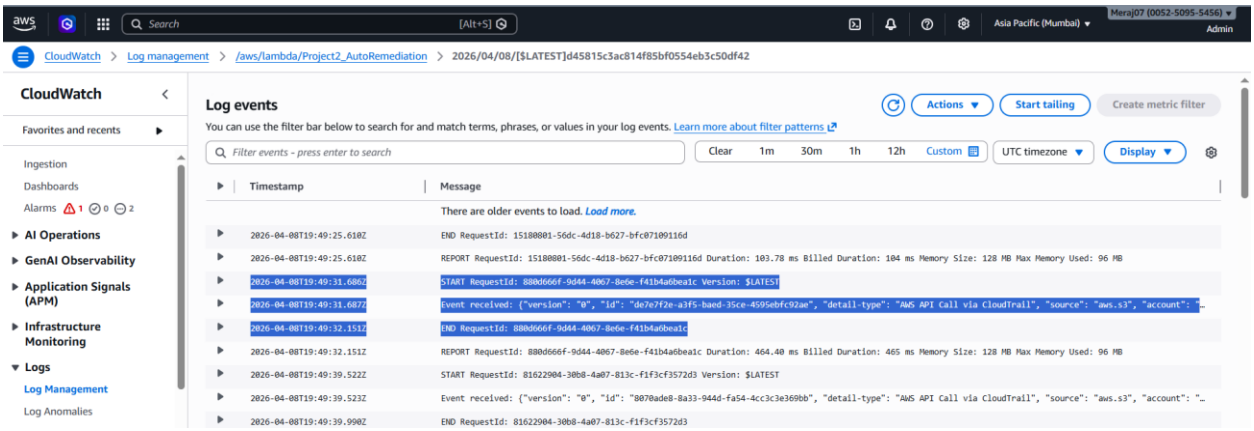


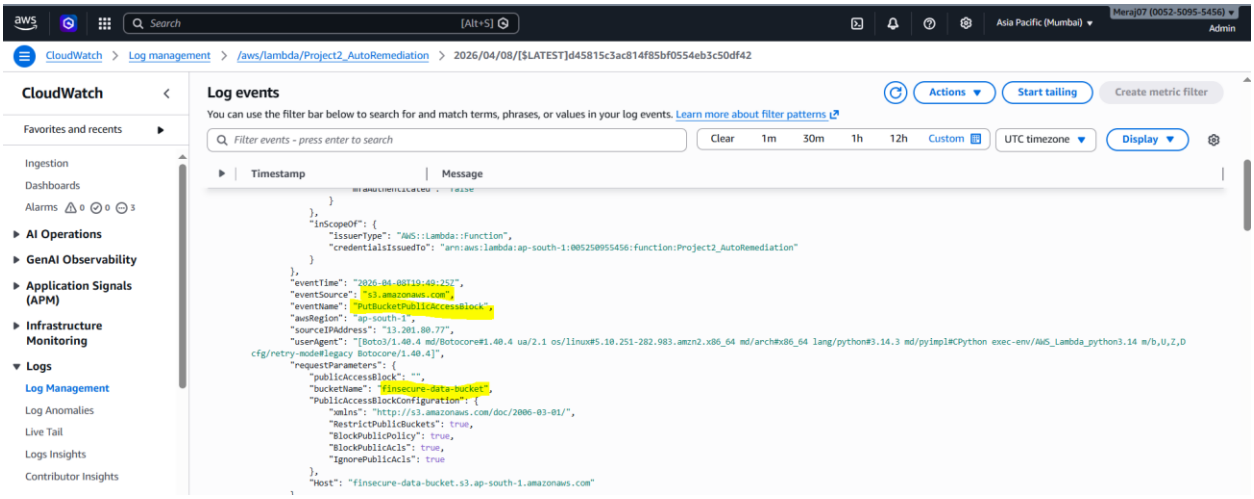


6. Lambda Auto-Remediation - Lambda function configured to automatically remediate security misconfigurations.

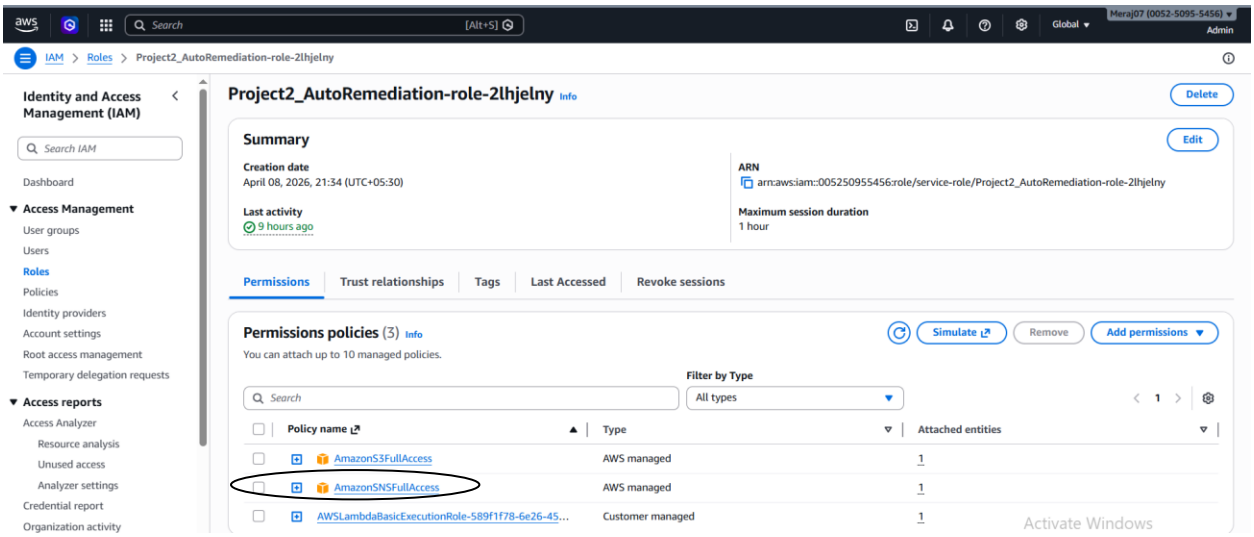


6.1 Lambda Auto-remediation execution logs - CloudWatch logs showing successful execution of auto-remediation actions

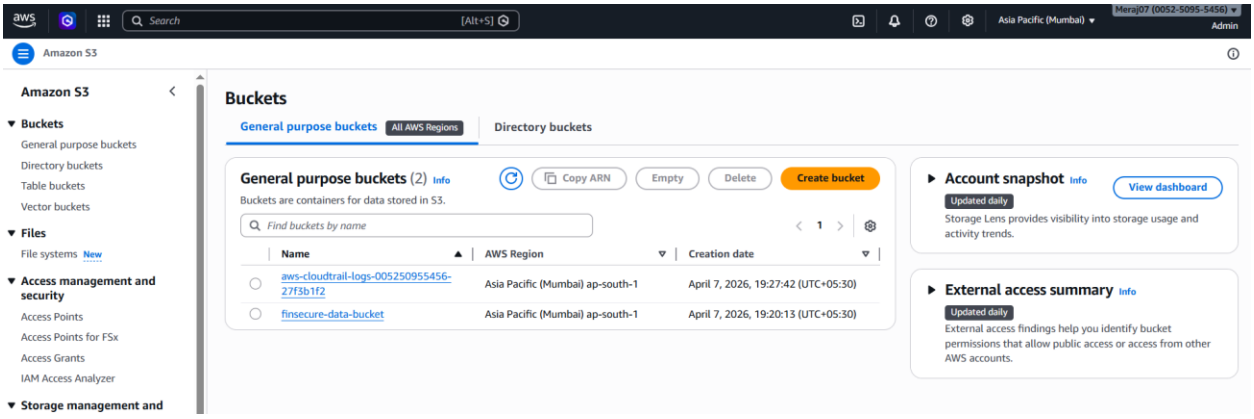




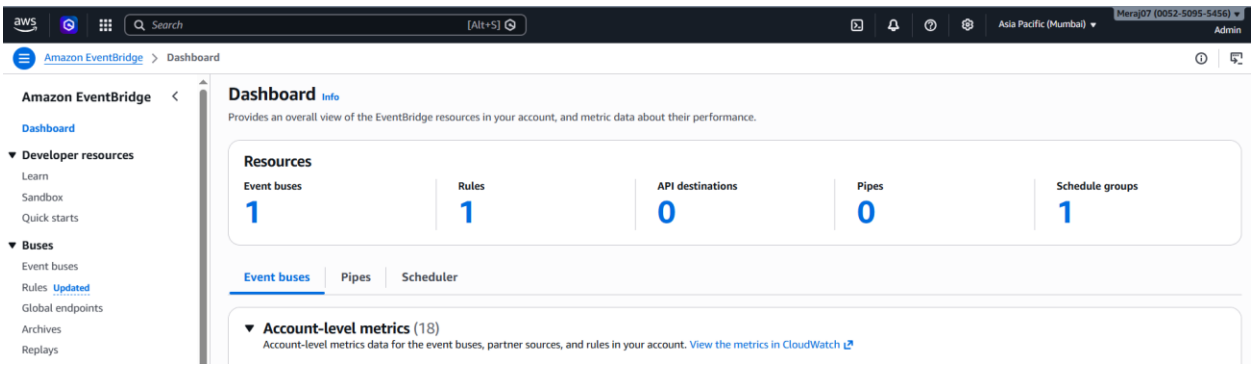
6.2 Lambda Role Permissions - Lambda execution role configured with least privilege permissions to update S3 and publish SNS alerts.



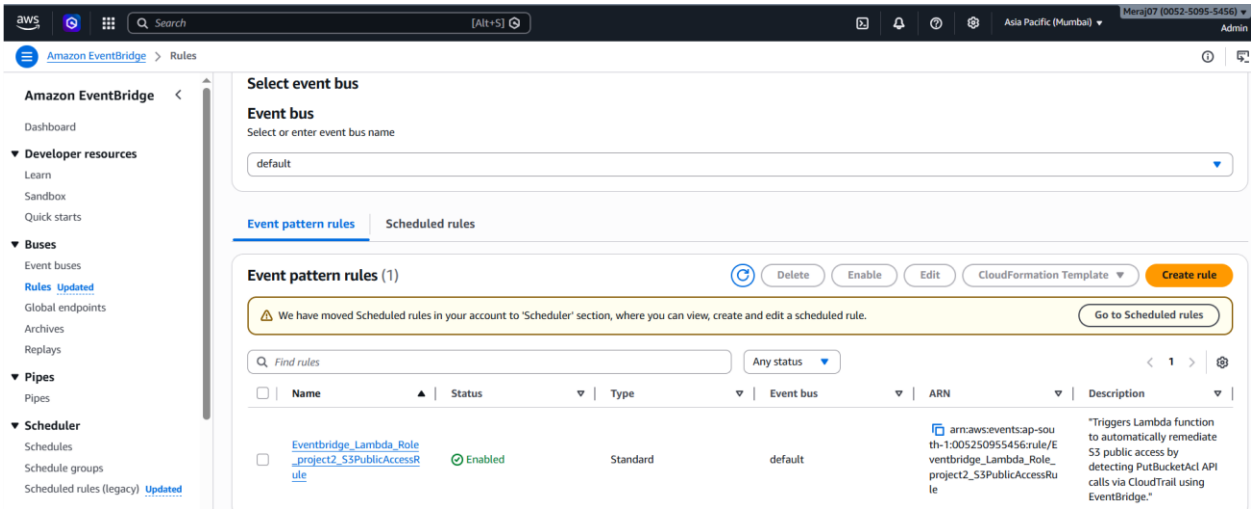
7. S3 Security Controls - S3 buckets configured with Block Public Access, versioning, and SSE-KMS encryption.

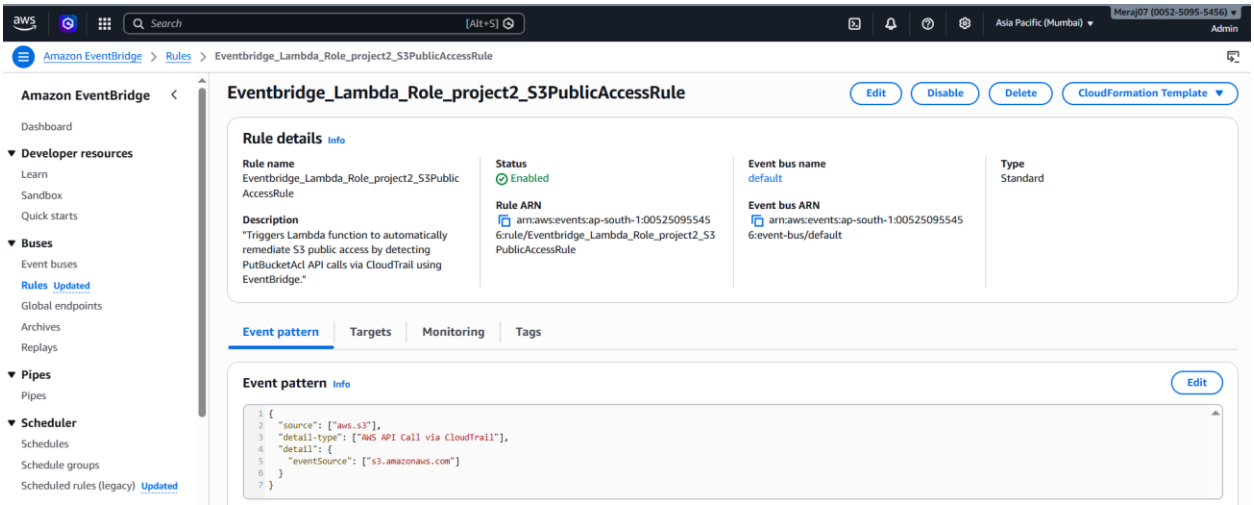


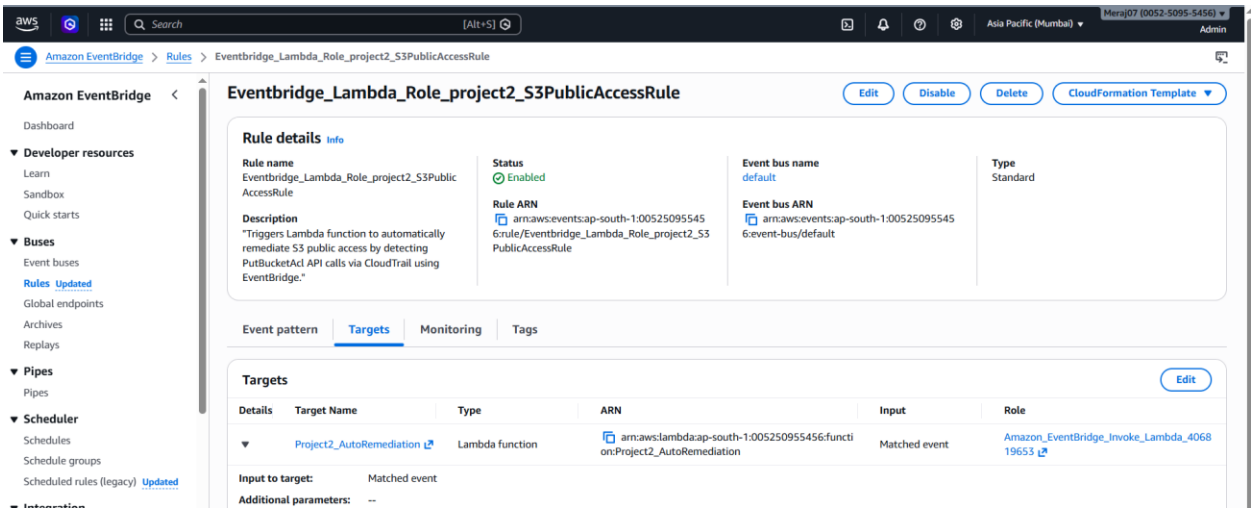
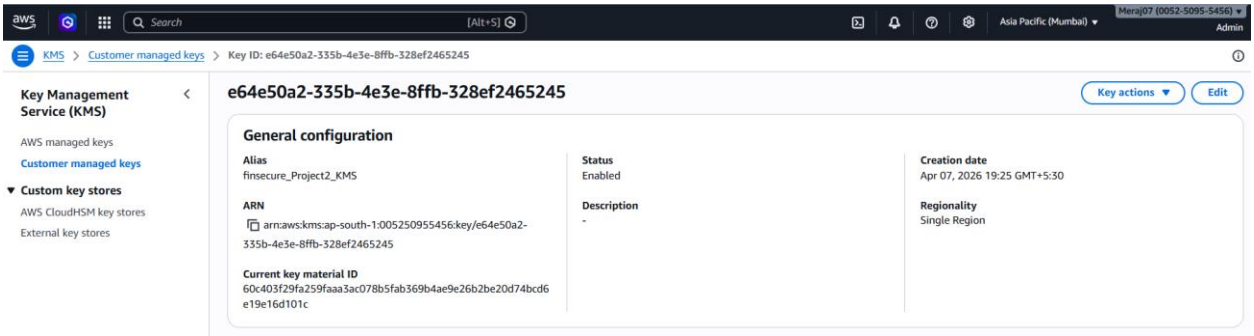
8. EventBridge Automation - EventBridge used to detect specific AWS API events and trigger Lambda for auto-remediation.



8.1 EventBridge Rule - Rule configured to detect S3 PutBucketAcl events and invoke Lambda function.







9. KMS Encryption - Customer-managed KMS key used for encrypting S3 buckets and CloudTrail logs with automatic key rotation enabled.